

The Substrate Is the Bottleneck: A JEPA-style Code World Model on Code Edits and Program Execution

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<https://github.com/AbdelStark/CodeLeWM>

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Abstract

Joint-embedding predictive architectures (JEPAs) learn by predicting future latent states rather than reconstructing observations. We ask whether a JEPA-style latent-transition recipe can learn useful dynamics over code. In CodeLeWM, the answer depends strongly on the substrate. Commit-message-conditioned Python edits over CommitPackFT-style data fail across four scaled Hugging Face Jobs runs: headline retrieval loses to the no-action baseline, mutation surprise is chance-level, and the v0.2 action-swap run collapses to an effective-rank ratio of 0.016. We then pivot the same model family and objective registry to input-conditioned execution triples, $(\text{code}, \text{input}) \mapsto \text{output}$, produced by a deterministic sandboxed Python executor. Across two v0.6 execution-substrate seeds, the no-action margin flips from -0.77 to +1.24, SIGReg drops by about 1200 $\times$, and the predicted-latent effective-rank ratio reaches 0.47. Evaluation on verified artifacts adds a narrow positive result: execution-pack retrieval beats no-action by +61.4 Recall@1 points and +65.9 MRR points on average, and semantic surprise decoy score gates reach AUC 1.0 on all configured decoy categories. The result is partial, not a broad code-generation or broad semantic-surprise claim: latent probes remain blocked by lexical controls, crash prediction is not evaluable, and HumanEval/MBPP-Plus reranking is not yet scored. The main lesson is that substrate signal quality dominates objective tuning.

1 Introduction

JEPA-style world models replace pixel or token reconstruction with prediction in a learned latent space. In images and video, this idea has produced self-supervised representations that avoid direct generative decoding while still learning useful structure [2, 6]. LeWorldModel extends the same latent-prediction view to end-to-end visual world models trained from pixels and actions [15]. Code is an intentionally hostile domain for that recipe. It is discrete, structured, sparse, and semantically brittle: one token can change behavior while large textual edits can be inert. The question is not just whether a JEPA objective can be ported to code, but which code substrate gives the objective enough signal to learn anything.

We study this question through a negative-to-positive substrate comparison in CodeLeWM, a latent world model for Python code transitions. In both settings, the model predicts the latent of an after-state from the latent of a before-state and an action:

$$\hat{z}_{\text{after}} = P(E(s_{\text{before}}), A(a)). \quad (1)$$

Both settings share the 256-dimensional transition contract, module family, and objective registry. This is not a perfect ablation: the v0.6 execution runner uses a substrate-specific loader, schedule, optimizer setting, and action-fusion surface. The comparison is therefore a substrate-pivot claim inside one implementation, not proof that every other variable is identical.

Substrate A treats a transition as a commit-message-conditioned code edit: before code plus a commit message maps to after code. It is attractive because it resembles the product surface of a patch scorer, but it has three structural problems. First, after $\approx$ before is often a strong prior. Second, commit messages are noisy, underspecified actions. Third, commits are often multi-purpose, so no single canonical action-conditioned transformation exists. The scaled evidence confirms the diagnosis. Four public runs all fail the headline retrieval comparison against no-action; the strongest v0.2 run reports an effective-rank ratio of 0.016, chance-level mutation surprise AUC 0.501, and latent probes beaten by lexical or metadata controls.

Substrate B keeps the latent-transition contract but changes the world. It treats a transition as a deterministic execution triple: $(\text{code}, \text{input}) \mapsto \text{output}$. Here the output is not a near-copy of the code, the input is a precise typed action, and each row is single-purpose. The v0.6 execution-substrate run inverts the failures above: two training seeds reach effective-rank ratios of 0.467 and 0.477, the no-action margin becomes strongly positive, and the execution-pack retrieval gate beats no-action by more than 60 absolute Recall@1 points.

The contribution is deliberately scoped:

1. A reproducible JEPA-style latent-transition recipe for code transitions, with manifest-backed data, checkpoints, and eval reports.
2. A negative result on commit-message-conditioned code edits, showing that objective tweaks do not rescue a low-signal substrate.
3. A partial-positive execution-substrate result under the same model family: non-collapse, action use, execution-pack retrieval, and semantic-decoy surprise score diagnostics pass across two seeds, while the broad semantic-surprise claim gate remains closed.
4. An explicit claim boundary: the result is not a code generator, not a general semantic-axis result, and not yet HumanEval/MBPP-Plus reranking evidence.

2 Related Work

Joint-embedding predictive architectures. I-JEPA predicts target image-block embeddings from context embeddings without pixel reconstruction [2]. V-JEPA extends this idea to video feature prediction [6]. LeWorldModel uses a latent prediction model for action-conditioned visual dynamics from pixels [15]. Our work preserves the same core contract but changes the domain to code and compares two substrates within one objective registry.

Anti-collapse regularization. The central technical risk in latent predictive models is collapse: a constant encoder can make prediction losses appear small. VICReg and Barlow Twins add variance, covariance, and redundancy-reduction constraints to self-supervised learning [5, 19]. LeJEPA and LeWorldModel study SIGReg-style regularization for stable joint-embedding learning [4, 15]. CodeLeWM inherits that anti-collapse emphasis and makes collapse gates a first-class artifact.

Code representations and code generation benchmarks. Pretrained code models such as CodeBERT, GraphCodeBERT, UniXcoder, and CodeT5 learn representations or generation models from program text and natural language [10–12, 17]. CodeNet, MBPP, APPS, HumanEval, and EvalPlus/MBPP-Plus provide benchmark substrates for program understanding, synthesis, and evaluation [3, 8, 13, 14, 16]. CodeT uses generated tests to rerank code generations [7]. CodeLeWM is not a generator: it is a local latent scorer/reranker that can be applied to candidate code or execution completions once labeled candidate artifacts exist.

Execution-aware program modeling. Execution has long been used as a supervised signal for neural program models, from synthetic program evaluation to source-code pretraining with execution traces [9, 18]. Recent execution-tuning work also studies models trained on program traces for output prediction and execution reasoning [1]. CodeLeWM differs in two ways: it predicts latent next-state embeddings rather than emitting traces, and its claim gate compares a commit-edit substrate against an execution substrate under one artifact discipline.

3 Model and Objective

Both substrates use the same transition-model contract. A state encoder E maps a bounded token capsule to $z \in \mathbb{R}^{256}$. An action encoder A maps either a commit message or an input representation to an action embedding. A predictor P maps $(z_{\text{before}}, A(a))$ to $\hat{z}_{\text{after}}$. The training loss is

$$\mathcal{L} = \|P(E(s_b), A(a)) - E(s_a)\|_2^2 + \lambda_{\text{sig}}\mathcal{L}_{\text{SIGReg}} + \lambda_{\text{swap}}\mathcal{L}_{\text{swap}} + \lambda_{\text{inv}}\mathcal{L}_{\text{inv}}. \quad (2)$$

The prediction term optimizes latent transition accuracy. SIGReg discourages collapse. The action-swap term penalizes predictions under mismatched actions, and the inverse-action term asks the model to preserve action information in the latent transition. The closest Substrate A comparand, the v0.2 action-swap run, and the v0.6 execution runs both enable this objective family. Their weights and training configs differ, so we interpret the result as evidence for a substrate pivot rather than as a single-factor architecture ablation.

4 Substrates

Substrate A: commit-message-conditioned edits. Substrate A uses permissively licensed CommitPackFT-style Python transitions. The before-state and after-state are code capsules; the action is a natural-language commit message. The v0.2 action-swap/inverse-action run packs 20,645 transitions with train/val/test counts 18,019/1,291/1,335. Its public report is the v0.2 action-swap HF results report; the run artifact id is listed in the claim audit.

Substrate B: input-conditioned execution. Substrate B uses deterministic Python execution records. The before-state is code, the action is a serialized input, and the after-state is a serialized output. The execution pack contains 1,605 records from CodeNet, MBPP, and APPS, with HumanEval and MBPP-Plus held out for downstream evaluation. Candidate source code is treated as untrusted input; sandbox execution is a data-build or operator-labeling step, not a model-training or model-scoring step. The v0.6 pack id is `codelewm-execution-pack-20260528T102625Z`; the report is `docs/benchmark/EXECUTION_V0_6_RESULTS_2026-05-30.md`.

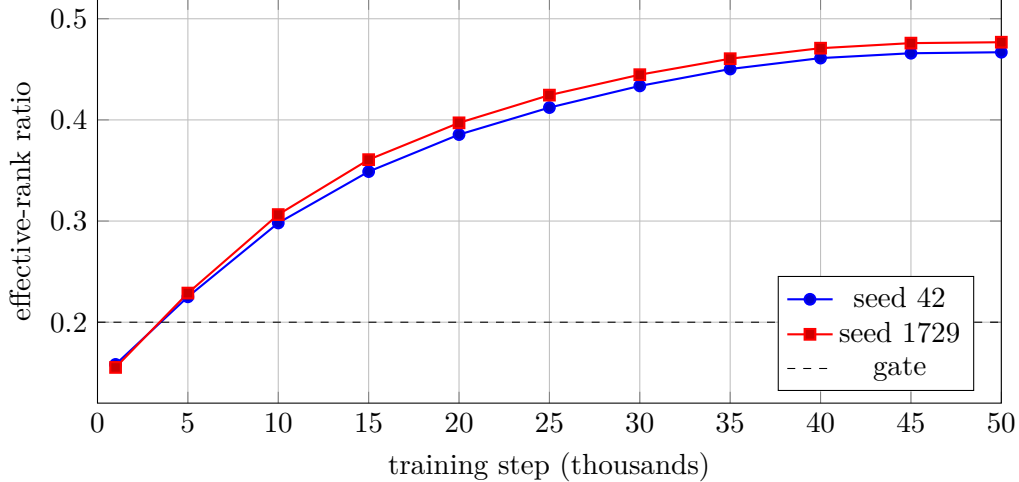


Figure 1: v0.6 predicted-latent effective-rank ratio from the collapse diagnostics report. Both seeds cross the 0.20 collapse gate early and finish near 0.47.

Table 1: Headline training and collapse metrics from the v0.2 and v0.6 benchmark reports. Raw MSE values are diagnostic within a substrate and should not be read as cross-substrate quality scores.

Run	prediction MSE	SIGReg	no-action margin	effective-rank ratio
v0.2 action-swap	0.1494	—	—	0.0158
v0.6 seed 42	0.00064	0.0364	+1.2308	0.4669
v0.6 seed 1729	0.00060	0.0348	+1.2434	0.4768

5 Experimental Protocol

All quantitative claims in this draft trace to checked-in benchmark reports or schema-versioned artifacts. For Substrate A, we use the v0.2 downloaded-artifact verification report. For Substrate B, we use two HF Jobs training seeds, 42 and 1729, each trained to 50k steps on `a10g-small1`. The #305 eval pass reruns the downloaded checkpoints through JSONL execution-pack eval surfaces: execution retrieval, semantic-decoy surprise, latent probes, and crash prediction. The committed eval tree is `docs/benchmark/v0_6/`; the same tree is mirrored to the HF dataset repo at the commit recorded in the claim audit, with the #322 semantic-surprise rerun recorded locally.

6 Results

6.1 Training dynamics and non-collapse

The training-time substrate-pivot claim passes on Substrate B and fails on Substrate A. Substrate A v0.2 reaches an effective-rank ratio of only 0.016. Substrate B reaches 0.467 and 0.477 on seeds 42 and 1729, above the configured 0.20 gate. Figure 1 shows the v0.6 predicted-latent effective-rank ratio over training.

The v0.6 no-action margin also flips sign quickly and remains positive in raw training metrics (Figure 2). The report-level final metrics average to +1.2308 for seed 42 and +1.2434 for seed 1729.

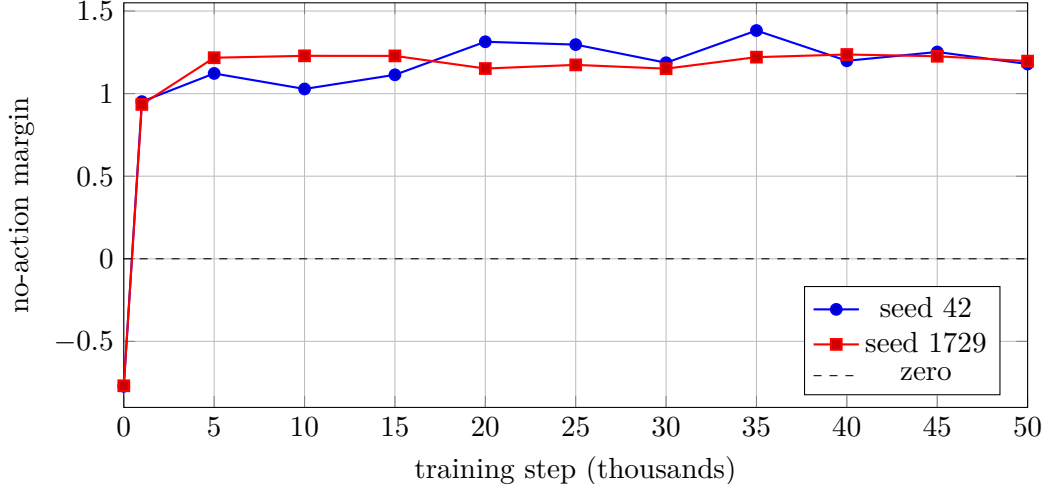


Figure 2: Raw v0.6 no-action margin training metric. Positive values mean the action-conditioned prediction is closer to the target than the no-action latent.

Table 2: Retrieval comparison. Positive deltas mean the action-conditioned model beats no-action.

Substrate	Run	Recall@1	Δ vs no-action	MRR	Δ vs no-action
A	#138	0.371	-0.088	0.473	-0.073
A	#154	0.363	-0.106	0.468	-0.082
A	#159	0.597	-0.053	0.675	-0.034
A	#172	0.263	-0.178	0.370	-0.163
B	v0.6 seed 42	0.657	+0.619	0.767	+0.663
B	v0.6 seed 1729	0.648	+0.610	0.759	+0.655

6.2 Retrieval

Retrieval is the cleanest action-use measurement. For Substrate A, the model must rank true after-states above hard negatives, but no-action remains stronger. For Substrate B, the query is an execution record and the candidate set is 236 val/test outputs. Table 2 shows the contrast.

Mean v0.6 Recall@1 is 0.6525 and mean MRR is 0.7628. Mean lift over no-action is +0.6144 Recall@1 and +0.6588 MRR, with seed-to-seed spread below one point. This supports a narrow execution-pack retrieval claim.

6.3 Surprise and decoys

Substrate A v0.2 mutation surprise is chance-level: mutation AUC 0.501. Substrate B v0.6 semantic surprise reruns pass all configured numeric score gates (Table 3). The caveat is important: same-problem-different submission still has only six scored pairs after filtering, so the explicit claim gate remains closed even though the broader semantic decoy pack contains 358 same-problem pairs across 68 problems.

Table 3: v0.6 semantic surprise AUCs and count-gate status from the committed per-seed surprise reruns.

Run	mutation	same-code/input	same-problem/submission	gate
v0.6 seed 42	1.000 / 236	1.000 / 352	1.000 / 6	closed
v0.6 seed 1729	1.000 / 236	1.000 / 352	1.000 / 6	closed

Table 4: v0.6 output-type latent-probe test accuracy.

Run	z_{pred}	no-action	lexical	claim
v0.6 seed 42	0.497	0.439	0.662	blocked
v0.6 seed 1729	0.599	0.541	0.618	blocked

6.4 Latent probes, crash prediction, and reranking

The broader downstream story is not positive. Only `output_type` has train/val/test labels for v0.6 latent probes. The latent view beats no-action by 5.7 points on both seeds, but lexical controls remain stronger (Table 4). The report therefore closes the positive representation claim gate.

Crash prediction is not evaluable because both v0.6 val/test slices have 0 positives and 236 negatives. HumanEval/MBPP-Plus reranking is also not scored: the sampler and `codelewm.eval.completion_label.v1` contract exist, but full live labeled-completion artifacts are not present. No HumanEval/MBPP-Plus pass@1 lift claim is allowed from the current evidence.

7 Discussion

The comparison supports one main explanation: the substrate is the bottleneck. On commit-message-conditioned edits, the model family has a strong shortcut. A no-action or identity-like prior is often correct, and the action text is too vague to force a different latent prediction. Objective variants improve some diagnostics but do not invert the headline gate.

Execution triples remove that shortcut. The output token distribution is different from the code distribution; copying the code latent is structurally wrong. The action is a deterministic value rather than a free-form commit message. Each row is single-purpose and judge-verifiable. Under these conditions, the execution objective produces non-collapsed latents and strong execution-pack retrieval.

This does not mean the model understands programs in a broad sense. The probe result is deliberately humbling: lexical features still beat the latent view on the only evaluable semantic target. The semantic-surprise score result is also limited by count gates: same-code/different-input decoys are plentiful, but same-problem/different-submission remains small-n. The right conclusion is not that CodeLeWM solves code generation; it is that a JEPA-style latent transition model can learn an action-conditioned execution substrate when the substrate supplies precise state/action/next-state signal.

8 Limitations

- The execution substrate is Python-only and stdlib-only under the current sandbox policy.

- The v0.6 result uses two training seeds. This is enough to check a large effect with small spread, but it is not a full variance study.
- The comparison is not a perfectly controlled ablation. The v0.6 execution run changes the loader, optimizer schedule, effective batch size, and action-fusion surface while preserving the latent-transition model family and objective registry. The causal claim is therefore scoped to the observed substrate pivot, not to substrate as the only changed variable.
- HumanEval and MBPP-Plus reranking are not evaluated in the current artifact set. A live OpenRouter sampling and hidden-test labeling run is still required.
- Generated surprise decoys are useful diagnostics but not full adversarial semantic tests. The strengthened semantic pack adds 352 same-code/different-input pairs, but the narrow same-problem/different submission category still has only six pairs.
- The paper compares substrates inside one implementation and objective registry. It does not claim superiority over large pretrained code models.

9 Reproducibility and Artifacts

The public artifact chain is intentionally part of the result. The code lives at <https://github.com/AbdelStark/CodeLeWM>. Substrate A uses `abdelstark/codelewm-public-shard`, `abdelstark/codelewm-transition-model`, and `abdelstark/codelewm-runs`. Substrate B uses `abdelstark/codelewm-execution-pack@v0.6.0` and the two seed-specific v0.6 run artifacts in `abdelstark/codelewm-runs`. The #305 eval reports are committed under `docs/benchmark/v0_6/` and mirrored to the HF dataset repo. Every published artifact is manifest-backed and secret-scanned.

The claim audit for this draft is `docs/papers/two_substrate_claim_audit.md`. The build command is:

```
scripts/build-two-substrate-paper
```

10 Conclusion

The negative v0.2 result and partial-positive v0.6 result are not contradictory. They are the point. A JEPA-style code world model does not become useful merely because the objective includes prediction, anti-collapse regularization, and action-contrastive terms. It needs a substrate where the action actually constrains the next state. For code, commit-message-conditioned edits did not provide that signal; deterministic execution triples did. The practical lesson for future code-world-model work is simple: pick the substrate before picking the architecture.

References

- [1] Jordi Armengol-Estap, Quentin Carbonneaux, Tianjun Zhang, Aram H. Markosyan, Volker Seeker, Chris Cummins, Melanie Kambadur, Michael F. P. O’Boyle, Sida Wang, Gabriel Synnaeve, and Hugh James Leather. What i cannot execute, i do not understand: Training and evaluating llms on program execution traces, 2025. URL <https://arxiv.org/abs/2503.05703>.

- [2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture, 2023. URL <https://arxiv.org/abs/2301.08243>.
- [3] Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models, 2021. URL <https://arxiv.org/abs/2108.07732>.
- [4] Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics, 2025. URL <https://arxiv.org/abs/2511.08544>.
- [5] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning, 2022. URL <https://arxiv.org/abs/2105.04906>.
- [6] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video, 2024. URL <https://arxiv.org/abs/2404.08471>.
- [7] Bei Chen, Fengji Zhang, Anh Nguyen, Daoguang Zan, Zeqi Lin, Jian-Guang Lou, and Weizhu Chen. Codet: Code generation with generated tests, 2022. URL <https://arxiv.org/abs/2207.10397>.
- [8] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code, 2021. URL <https://arxiv.org/abs/2107.03374>.
- [9] Yangruibo Ding, Ben Steenhoeck, Kexin Pei, Gail Kaiser, Wei Le, and Baishakhi Ray. Traced: Execution-aware pre-training for source code, 2023. URL <https://arxiv.org/abs/2306.07487>.
- [10] Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, Ming Gong, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, and Ming Zhou. Codebert: A pre-trained model for programming and natural languages, 2020. URL <https://arxiv.org/abs/2002.08155>.
- [11] Daya Guo, Shuo Ren, Shuai Lu, Zhangyin Feng, Duyu Tang, Shujie Liu, Long Zhou, Nan Duan, Alexey Svyatkovskiy, Shengyu Fu, Michele Tufano, Shao Kun Deng, Colin Clement, Dawn Drain, Neel Sundaresan, Jian Yin, Daxin Jiang, and Ming Zhou. Graphcodebert: Pre-training code representations with data flow, 2021. URL <https://arxiv.org/abs/2009.08366>.
- [12] Daya Guo, Shuai Lu, Nan Duan, Yanlin Wang, Ming Zhou, and Jian Yin. Unixcoder: Unified cross-modal pre-training for code representation, 2022. URL <https://arxiv.org/abs/2203.03850>.

- [13] Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, and Jacob Steinhardt. Measuring coding challenge competence with apps, 2021. URL <https://arxiv.org/abs/2105.09938>.
- [14] Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by chatgpt really correct? rigorous evaluation of large language models for code generation, 2023. URL <https://arxiv.org/abs/2305.01210>.
- [15] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworld-model: Stable end-to-end joint-embedding predictive architecture from pixels, 2026. URL <https://arxiv.org/abs/2603.19312>.
- [16] Ruchir Puri, David S. Kung, Geert Janssen, Wei Zhang, Giacomo Domeniconi, Vladimir Zolotov, Julian Dolby, Jie Chen, Mihir Choudhury, Lindsey Decker, Veronika Thost, Luca Buratti, Saurabh Pujar, Shyam Ramji, Ulrich Finkler, Susan Malaika, and Frederick Reiss. Codenet: A large-scale ai for code dataset for learning a diversity of coding tasks, 2021. URL <https://arxiv.org/abs/2105.12655>.
- [17] Yue Wang, Weishi Wang, Shafiq Joty, and Steven C. H. Hoi. Codet5: Identifier-aware unified pre-trained encoder-decoder models for code understanding and generation, 2021. URL <https://arxiv.org/abs/2109.00859>.
- [18] Wojciech Zaremba and Ilya Sutskever. Learning to execute, 2015. URL <https://arxiv.org/abs/1410.4615>.
- [19] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction, 2021. URL <https://arxiv.org/abs/2103.03230>.